Practice workbook · Define a gene mutation and name its causes

Stating what a mutation is, how it alters a chromosome, and naming mutagens or why a tissue such as skin is prone.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-MUTATION-AND-CAUSES

1. Question 5 · part a.i · 2 marks

November 2024 Paper 2

Worked-example retry · Attempt part a.i only.

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5 (a) (i) Describe what is meant by the term **mutation**.

(2)

(ii) Explain how a mutation can produce a change in the characteristics of an organism.

(3)

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1. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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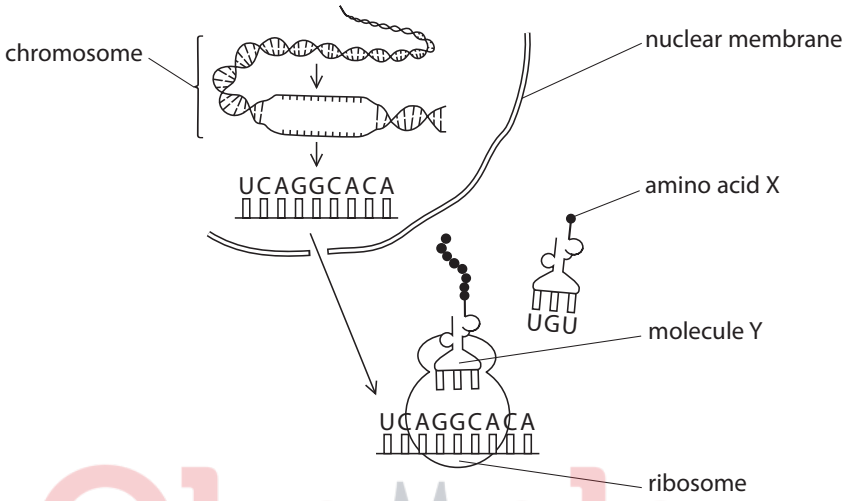
2. Question 6 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

6 Protein synthesis has a number of stages.

The diagram shows some of the stages.



The table lists the base triplets of DNA that code for some of the amino acids found in a protein.

Base triplet of DNA	Amino acid
AGT	serine
CCG	glycine
TGT	threonine
GTA	histidine
CAA	valine
TCC	argenine
ACA	cysteine

(a) Describe the function of the nuclear membrane.

(2)

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2. Question 6 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

(b) Which is the correct sequence for the three unpaired bases on the tRNA molecule labelled Y?

(1)

- ☒ **A** AAU
- ☒ **B** CCG
- ☒ **C** GGC
- ☒ **D** UUA

(c) Give the name of amino acid X.

(1)

(d) Describe the process occurring on the ribosome.

(6)

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2. Question 6 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

(e) Describe how a mutation will affect a chromosome.

(2)

(Total for Question 6 = 12 marks)

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Working and answer

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3. Question 8 · part a · 3 marks

November 2020 Paper 2

Further practice · Attempt part a only.

8 Haemophilia is a sex-linked condition caused by a recessive allele X^h instead of the usual X^H.

Haemophilia is the result of a mutation.

(a) Explain what is meant by the term **mutation**.

(3)

(b) (i) A man without haemophilia and a woman who is a carrier of haemophilia have a child.

Draw a genetic diagram to show the possible genotypes and phenotypes of this child.

(4)

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3. Question 8 · part a · 3 marks · Working and answer

Working and answer

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4. Question 3 · part a · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a only.

3 (a) Polycystic kidney disease (PKD) is an inherited condition caused by a gene mutation.

PKD affects the function of the kidneys.

State what is meant by a gene mutation.

(1)

.....

.....

(b) PKD occurs when a child inherits a faulty allele from one parent.

One parent has a genotype Hh and the other parent has a genotype hh.

The parents have four children.

Draw a genetic diagram to show the possible genotypes of the four children.

(2)

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(c) Some people with PKD need a kidney transplant.

Give one advantage and one disadvantage of a kidney transplant.

(2)

Advantage

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Disadvantage

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(Total for Question 3 = 5 marks)

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4. Question 3 · part a · 1 mark · Working and answer

Working and answer

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5. Question 5 · part b.ii · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.ii only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

5. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

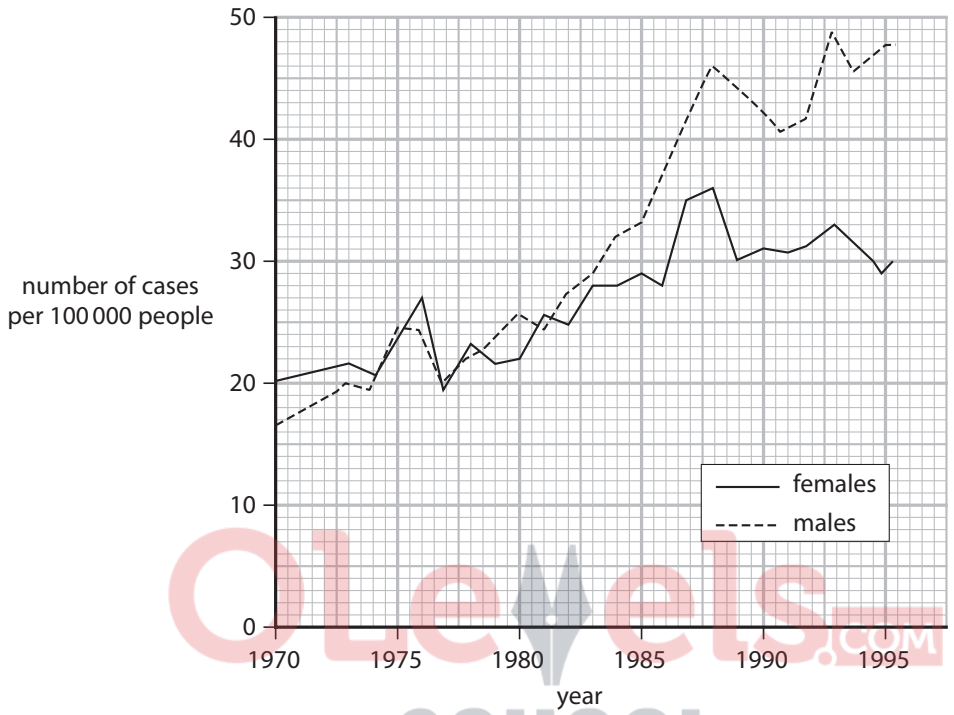
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6. Question 6 · part a.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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6. Question 6 · part a.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

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number of females with melanoma = ...

(Total for Question 6 = 11 marks)

6. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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